NON-OVERLAPPING ENSEMBLE WEIGHTING VIA MÖBIUS DECOMPOSITION: A COOPETITIVE FRAMEWORK FOR DIAGNOSING AND LEVERAGING MODEL INTERACTIONS

A Special Problem Presented to the

Faculty of the Department of Computer Science,

University of the Philippines Cebu

In Partial Fulfillment

Of the Requirements for the Degree

Bachelor of Science in Computer Science

MAXELL G. MILAY

Student

ASST. PROF. DHARYLL PRINCE M. ABELLANA

Special Problem Adviser

June 2026

UNIVERSITY OF THE PHILIPPINES

College of Science

Department of Computer Science

ACCESS PERMISSION

Non-Overlapping Ensemble Weighting via Möbius Decomposition: A Coopetitive Framework for Diagnosing and Leveraging Model Interactions

Permission is given for the following people to have access to this SP:

Available to the general public	Yes
Available only after consultation with author/SP adviser	Yes
Available only to those bound by confidentiality agreement	Yes

---

MAXELL G. MILAY

Student

ASST. PROF. DHARYLL PRINCE M. ABELLANA

Special Problem Adviser

APPROVAL SHEET

The special problem entitled: **"Non-Overlapping Ensemble Weighting via Möbius Decomposition: A Coopetitive Framework for Diagnosing and Leveraging Model Interactions"** submitted and prepared by MAXELL G. MILAY is approved by:

____

DHARYLL PRINCE M. ABELLANA

Special Problem Adviser

Date Signed: ____________________

____

ELI EDRIAN L. TAN

Officer-in-Charge

Department of Computer Science

Date Signed: ____________

____

DR. ALVIN G. ROXAS

Dean

College of Science

Date Signed: ____________

ABSTRACT

When combining multiple machine learning models into an ensemble, one of the most persistent challenges is deciding how much weight to assign each model. The most principled existing approach, based on cooperative game theory, produces a single score per model that mixes together how well a model performs on its own and how well it works alongside others – making it impossible to tell which source of value is actually driving the weight. Attempts to fix this by adding a separate interaction measure on top create a new problem: the interaction effects get counted twice, distorting the weights and degrading predictive accuracy. This study introduces a coopetitive ensemble weighting framework that resolves both problems using the Möbius decomposition, a mathematical technique that cleanly separates each model's individual contribution from its pairwise interaction effects with no overlap. A tunable parameter controls how much influence model interactions have on the final weights, and the framework produces a pairwise dividend matrix that maps, for every pair of models in the pool, whether they complement or duplicate each other.

The framework was evaluated on fifteen real-world binary classification datasets using five diverse base models. The proposed approach achieved the highest discriminative accuracy and the best probability calibration among all tested configurations, and the double-counting variant performed measurably worse – confirming that the theoretical concern has practical consequences. While the improvement over simpler baselines was modest and did not reach statistical significance in this study, the most important contribution is the pairwise dividend matrix as a diagnostic tool for improving predictive performance. A strongly negative matrix signals that the model pool is dominated by redundancy, pointing practitioners toward replacing or removing overlapping models as the most direct path to higher accuracy. By linking what the matrix reveals about model interactions to concrete corrective actions, this framework offers a level of interpretable, actionable guidance that no other ensemble weighting method currently provides.

ACKNOWLEDGEMENTS

The completion of this research was made possible through the collective support, guidance, and generosity of individuals and institutions whose contributions I wish to formally recognize.

First, I give thanks to God, from whom all the capacity for inquiry, perseverance, and understanding ultimately derives.

To my special problem adviser, Asst. Prof. Dharyll Prince M. Abellana, I extend my sincere gratitude for the guidance and direction you provided throughout this work. It is a testament to the quality of your mentorship that this paper found its completion despite the challenges along the way.

To my family, whose silent sacrifices have supported every academic endeavor I have pursued, I owe more than words can adequately convey. The foundation you built made this work possible.

To my friends who stood by me through the most challenging phases of this research, offering both thoughtful engagement and personal encouragement when the demands of the work felt overwhelming, thank you for your presence and your patience.

To the University of the Philippines Cebu, thank you for fostering an academic environment that values undergraduate research and gives students the space to pursue work that is both original and technically ambitious.

TABLE OF CONTENTS

List of Figures

List of Tables

CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Rationale

Ensemble methods represent one of the most effective strategies in machine learning for improving predictive performance by combining multiple base models. Despite their widespread adoption, a persistent and fundamental challenge remains: determining how much weight each constituent model should receive within the ensemble. Simple approaches such as equal weighting or performance-proportional weighting fail to account for the pairwise and higher-order interdependencies among models, often resulting in suboptimal combinations where redundant models dilute the contributions of complementary ones (Kuncheva & Whitaker, 2003).

The Shapley value from cooperative game theory (Shapley, 1953) offers a principled alternative by assigning each model a weight proportional to its average marginal contribution across all possible sub-ensembles. Recent work has established the promise of Shapley-based approaches in ensemble weighting and model valuation (Rozemberczki & Sarkar, 2021; Rozemberczki et al., 2022; Bimonte et al., 2024). However, the Shapley value has a structural limitation: because it averages each model's contribution over all coalition contexts, it entangles each model's standalone quality with its interaction effects relative to the other models in the pool. A model with modest standalone performance but strong complementarity with others may receive the same Shapley weight as one with high standalone performance but redundant interactions, with no way to distinguish the two cases from the Shapley weight alone. Practitioners who attempt to correct for this entanglement by augmenting Shapley values with the Shapley Interaction Index (SII) inadvertently introduce double-counting: the interaction information already embedded in the Shapley value is then attributed a second time through the additive SII terms, distorting the resulting weights. Three specific technical gaps arising from these limitations are formalized in § 1.2.

This study proposes a framework that resolves these gaps using the Möbius decomposition, also known as Harsanyi dividends (Harsanyi, 1959, 1963). The Möbius transform uniquely decomposes the characteristic function of the ensemble game into non-overlapping contributions: singleton dividends that capture each model's pure standalone value and pairwise dividends that capture the surplus or deficit arising exclusively from model interactions (Dehez, 2017). The resulting pairwise dividend matrix serves as an interpretable diagnostic, revealing which model pairs are complementary, redundant, or independent.

Drawing on the concept of coopetition (Brandenburger & Nalebuff, 1996), the framework is termed "coopetitive" because the cooperative component (singleton dividends measuring standalone contribution) and the competitive component (negative pairwise dividends penalizing redundancy) are balanced through a tunable parameter λ. The proposed weighting formula is deliberately simple; its value lies not in computational sophistication but in providing a principled, interpretable mechanism for diagnosing and leveraging model interactions that more powerful but opaque methods like stacking do not offer. While the Shapley Interaction Index has been used extensively for feature-level attribution in explainable AI (Tsai et al., 2023; Muschalik et al., 2024), no study identified in the literature review applied the Möbius decomposition to construct ensemble weights that cleanly separate individual and interaction contributions (established through the literature review in § 2.3). This study addresses that gap.

1.2 Statement of the Problem

The three gaps introduced in § 1.1 are articulated formally below. The first is the entanglement gap. The Shapley value (Shapley, 1953; Rozemberczki et al., 2022), the most theoretically grounded approach to model valuation in ensembles, entangles individual and interaction effects into a single aggregate score. Specifically, because

$$\phi_i = \sum_{T \ni i} \frac{m(T)}{|T|}, \tag{1.1}$$

the Shapley value of model i incorporates diluted versions of all interaction dividends involving i (where $m(T)$ denotes the Harsanyi dividend of subset T, defined formally in § 2.3). This means that a model with modest standalone performance but strong complementarity with others may receive the same Shapley weight as one with high standalone performance but redundant interactions. The inability to disentangle these two dimensions prevents practitioners from understanding why a model receives a particular weight and limits the potential for targeted ensemble optimization.

The second is the double-counting gap. If practitioners attempt to correct for interactions by augmenting Shapley values with the Shapley Interaction Index, the resulting formulation

$$s_i = \phi_i + \lambda \sum_{j \neq i} \Delta_{ij} \tag{1.2}$$

(where $\lambda > 0$ is a scalar weight) introduces systematic double-counting of pairwise effects. The interaction information already embedded within ϕ_i is counted again through the additive Δ_{ij} terms, distorting the resulting weights and undermining the theoretical guarantees of the approach. To the author's knowledge, no established ensemble weighting method currently addresses this double-counting problem. (A systematic review of the ensemble weighting literature, including the surveys of Rozemberczki et al. 2022 and Kuncheva & Whitaker 2003, found no method that explicitly identifies or corrects for this issue.)

The third is the interpretability gap. Existing ensemble combination methods, including stacking and blending, optimize weights through cross-validated meta-learning. While individual model quality can often be assessed in isolation (for example, through standalone accuracy or the model's own Shapley value in the ensemble game), no standard ensemble method produces a pairwise diagnostic that reveals which specific model pairs are complementary, redundant, or independent within a given pool. Practitioners cannot determine from the ensemble weights alone how the interaction structure of the pool influenced the final combination, limiting the ability to diagnose ensemble behavior, guide model selection, and communicate results to stakeholders.

The Möbius decomposition directly resolves all three gaps by construction – meaning the non-overlapping property is an algebraic identity that follows directly from the definition of the dividends, not an empirical claim – as established in § 2.3.

1.3 Research Objectives

The primary objective of this research was to design and empirically evaluate a coopetitive ensembling framework that uses the Möbius decomposition to construct ensemble weights from non-overlapping individual and interaction components. This overarching objective was operationalized through the following specific aims.

The first aim was to design a coopetitive weight formula that combines singleton Harsanyi dividends $m(\{i\})$, capturing each model's pure standalone contribution, with pairwise dividends $m(\{i, j\})$, capturing interaction effects, through a tunable parameter λ that controls the trade-off between individual model quality and interaction structure. The formula (defined in § 1.6) ensures non-overlapping accounting of individual and interaction information.

The second aim was to establish the construct validity of pairwise Harsanyi dividends as measures of model complementarity and redundancy through controlled experiments involving synthetic redundancy injection, monotonicity testing, correlation with established diversity metrics, and negative control conditions.

The third aim was to conduct a systematic ablation study that isolates the contribution of each component (individual dividends, interaction dividends, and their combination) by comparing six primary variants (equal, performance-weighted, individual-only ($\lambda = 0$), interaction-only, coopetitive with fixed λ, and coopetitive with cross-validated λ) together with supplementary Shapley-based variants across multiple benchmark datasets. Because the individual-only variant ($\lambda = 0$) is algebraically equivalent to performance-weighted weighting when AUC is the metric, this equivalence first serves as a sanity check. The ablation then tests whether the interaction term ($\lambda > 0$) adds value beyond this established baseline.

The fourth aim was to benchmark the coopetitive framework against established ensemble methods, including equal weighting, performance-weighted weighting, cooperative Shapley weighting, stacking with cross-validated logistic regression, and the best single model, using Friedman tests with Holm-corrected pairwise Wilcoxon post-hoc comparisons.

The fifth aim was to characterize the full sensitivity landscape of ensemble performance to the λ parameter through a systematic sweep across $\lambda \in [0, 2.0]$, complementing the binary question of Aim 3 (whether $\lambda > 0$ helps) by mapping the shape and dataset-dependence of the performance–parameter curve, and to determine whether a fixed default value can achieve near-optimal performance across datasets.

1.4 Significance of the Study

This research holds significance for both the theoretical foundations of ensemble learning and the practical application of model combination strategies.

From a theoretical perspective, this study contributes to the body of knowledge by providing the first empirical evaluation of Möbius-based ensemble weighting that cleanly separates individual model quality from interaction structure. While the Shapley Interaction Index has been extensively studied in the context of feature attribution and explainable AI (Tsai et al., 2023; Muschalik et al., 2024), its application to ensemble weight construction has remained unexplored. By demonstrating that the non-overlapping property of Harsanyi dividends can serve as a principled foundation for ensemble combination, this study extends cooperative game theory's applicability within machine learning. The construct validity experiments characterize these relationships empirically; detailed findings are presented in § 4.1.2.

From a practical perspective, this research offers several contributions to practitioners and tool developers. For practitioners in data science and machine learning, the framework provides an interpretable diagnostic tool in the form of the pairwise dividend matrix, which reveals which model pairs are complementary, redundant, or independent. This information directly informs decisions about model selection, pool construction, and ensemble design. For researchers working on automated machine learning and ensemble selection, the coopetitive weight formula offers a computationally tractable alternative to stacking that does not require cross-validated meta-learning, making it suitable for scenarios where computational budgets are limited or where interpretability is a priority. For the broader machine learning community, the λ sensitivity analysis provides practical guidance on default parameter selection, and the systematic comparison with established methods clarifies the conditions under which interaction-aware weighting offers meaningful improvements over simpler baselines.

1.5 Scope and Delimitation

This study evaluated the framework using a pool of five heterogeneous base models (Logistic Regression, Random Forest, SVM, XGBoost, and k-Nearest Neighbors) on fifteen binary classification datasets from the OpenML and UCI repositories, using AUC-ROC as the primary metric and negative Brier score as a secondary check. The Möbius decomposition was truncated to singleton and pairwise terms only; higher-order interactions, multi-class tasks, regression, and real-time deployment scenarios were outside the scope of this study. Exact coalition evaluation was used, limiting the framework to pools of $n \leq 12$ models. Full methodological details, including the data-splitting protocol and delimitations, are specified in Chapter 3.

1.6 Definition of Terms

To ensure clarity and a shared understanding of the concepts discussed in this study, the following terms are defined as they are used within this research context.

Characteristic function, denoted $v(S)$, refers to a function that maps each coalition S of base models to a performance score, specifically AUC-ROC, measured on a held-out validation set. The characteristic function forms the foundation of the cooperative game formulation that underlies the entire framework.

Möbius decomposition, also known as Harsanyi dividends, refers to the unique decomposition of the characteristic function $v(S)$ into non-overlapping contributions from subsets of models. Formally, $v(S) = \sum_{T \subseteq S} m(T)$, where each $m(T)$ is computed via the inclusion–exclusion formula $m(T) = \sum_{R \subseteq T} (-1)^{|T|-|R|} v(R)$.

Singleton dividend, denoted $m(\{i\})$, refers to the Harsanyi dividend for a single model i, computed as $m(\{i\}) = v(\{i\}) - v(\emptyset)$. This quantity captures the model's pure standalone contribution to the ensemble, with no interaction effects mixed in.

Pairwise dividend, denoted $m(\{i, j\})$, refers to the Harsanyi dividend for a pair of models, computed as $m(\{i, j\}) = v(\{i, j\}) - v(\{i\}) - v(\{j\}) + v(\emptyset)$. A positive pairwise dividend indicates complementarity between the two models, meaning their joint contribution exceeds the sum of their individual contributions. A negative dividend indicates redundancy, meaning the models overlap in predictive signal.

Coopetitive score, denoted s_i, refers to the combined score for model i computed as $s_i = m(\{i\}) + \lambda \frac{1}{2} \sum_{j \neq i} m(\{i,j\})$, where λ is the interaction weight parameter. This score integrates both individual quality and interaction structure into a single value from which the final ensemble weight is derived. Final ensemble weights are derived from s_i via softmax normalization with an adaptive temperature parameter; see § 3.2.5.

Interaction weight λ refers to the tunable parameter that controls the trade-off between individual model quality and interaction structure in the coopetitive score. When $\lambda = 0$, the framework reduces to pure performance-weighted weighting; as λ increases, the weights become more sensitive to the complementarity and redundancy structure of the model pool.

Construct validity refers to the degree to which the pairwise Harsanyi dividends accurately measure the theoretical constructs of model complementarity and redundancy. This was established through controlled experiments with known ground truth, including synthetic redundancy injection and negative control conditions.

Shapley Interaction Index, denoted Δ_{ij}, refers to the coalition-averaged measure of pairwise interaction between models i and j, as defined by Grabisch & Roubens (1999). Unlike the pairwise Harsanyi dividend, which evaluates interaction at the empty coalition only, the SII averages interactions across all possible coalition contexts. The context-sensitivity diagnostic compared these two measures.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Ensemble Methods and the Weighting Problem

Ensemble methods have long been recognized as one of the most effective strategies for improving predictive performance in machine learning. The fundamental principle underlying all ensemble approaches is that combining diverse models can reduce variance, bias, or both, yielding predictions that are superior to any individual constituent model (Kuncheva & Whitaker, 2003). Early approaches such as bagging (Breiman, 1996) and boosting (Freund & Schapire, 1997) demonstrated that even simple combination rules could yield substantial performance gains, and subsequent developments in stacking (Wolpert, 1992) and blending introduced the idea of learning an optimal combination through a meta-learner trained on base model outputs.

Despite this progress, the question of how to optimally weight base models within an ensemble remains an active and unresolved area of research. The simplest strategy, equal weighting, treats all models as interchangeable regardless of their individual quality or their relationships with other models in the pool. Performance-proportional weighting assigns higher weights to models with better standalone accuracy, but this approach ignores the interaction structure of the ensemble entirely. A model with moderate standalone performance but strong complementarity with others will be systematically underweighted (Kuncheva & Whitaker, 2003). Stacking with cross-validated meta-learners (Wolpert, 1992) can implicitly capture some interaction effects, but the resulting weights are opaque and provide no interpretable information about model relationships: the meta-learner coefficients reflect how to combine outputs, not which model pairs are complementary or redundant. Ensemble pruning methods (e.g., early work on pruning AdaBoost by Margineantu & Dietterich 1997), which address the complementary problem of pool composition by selecting a subset of models rather than weighting all of them, offer a related perspective; unlike weighting, pruning methods select a complementary subset of models rather than assigning graded weights to all of them. The pairwise dividend matrix developed in this study provides exactly the information needed to make such pruning decisions in a principled way, bridging the gap between weighting and selection. This gap between the effectiveness of ensemble methods and the principled understanding of their combination dynamics motivates the application of game-theoretic concepts to ensemble weighting.

2.2 Cooperative Game Theory in Machine Learning

Cooperative game theory provides a natural mathematical framework for reasoning about the contributions of individual agents within a collective. The Shapley value (Shapley, 1953), the most widely studied solution concept, assigns each player a payoff proportional to their average marginal contribution across all possible coalitions. In the context of ensemble learning, each base model is treated as a player, and the characteristic function $v(S)$ maps each coalition of models to a performance metric evaluated on a held-out set (Rozemberczki & Sarkar, 2021).

Recent work has established the applicability of Shapley values to ensemble problems. Rozemberczki & Sarkar (2021) formalized the ensemble weighting problem as a cooperative game and showed that Shapley-based weights could outperform equal weighting in several settings. Rozemberczki et al. (2022) extended this line of work by surveying the broader applications of the Shapley value in machine learning, including data valuation, feature attribution, and model selection. Bimonte et al. (2024) applied Shapley-based ensemble weighting to mortality forecasting models, demonstrating that cooperative game formulations could improve predictive accuracy in actuarial applications.

However, the standard Shapley value has a structural property that limits its usefulness for ensemble weight construction. As established by the Möbius representation (Harsanyi, 1959; Dehez, 2017), the Shapley value can be written as $\phi_i = \sum_{T \ni i} m(T)/|T|$, where $m(T)$ are the Harsanyi dividends. This expression reveals that the Shapley value inherently mixes a model's standalone contribution $m(\{i\})$ with all higher-order interaction dividends involving model i, each diluted by the size of the respective subset. Consequently, practitioners cannot determine from the Shapley value alone whether a model's high weight is due to its strong standalone performance, its complementarity with other models, or some combination of both. This entanglement motivates the search for decomposition-based approaches that separate these two sources of value. The double-counting problem that arises when practitioners attempt this correction is formalized in § 1.2 and addressed empirically in § 4.2.3.

2.3 The Möbius Decomposition and Harsanyi Dividends

The Möbius decomposition, introduced in the context of cooperative games as Harsanyi dividends (Harsanyi, 1959, 1963), provides a unique decomposition of any characteristic function $v(S)$ into non-overlapping contributions from every subset of players. Theorem 1 in Dehez (2017) states that for any characteristic function v, there exists a unique collection of dividends $\{m(T)\}_{T\subseteq N}$ satisfying $v(S) = \sum_{T\subseteq S} m(T)$ for every $S \subseteq N$, computed by the inclusion–exclusion formula. Each coalition's surplus is thus assigned to exactly one subset T, with no term appearing in more than one place. As displayed equations:

$$v(S) = \sum_{T\subseteq S} m(T), \tag{2.1}$$

$$m(T) = \sum_{R\subseteq T} (-1)^{|T|-|R|} v(R). \tag{2.2}$$

The Möbius decomposition has been applied in several domains beyond ensemble learning. In multicriteria decision making, it serves as the canonical representation of fuzzy measures (Choquet capacities), where singleton and pairwise dividends correspond to individual criterion importance and synergy between criteria (Grabisch & Roubens, 1999). In explainable AI, higher-order Harsanyi dividends have been used to define interaction-aware feature attribution methods (Tsai et al., 2023; Muschalik et al., 2024). In cooperative game theory itself, the dividends underlie the definition of the Shapley value and related solution concepts (Dehez, 2017). However, none of these applications construct ensemble weights that explicitly separate individual model quality from pairwise interaction effects. The Shapley-based ensemble methods of Rozemberczki & Sarkar (2021) and Bimonte et al. (2024) use the aggregate Shapley value, which entangles these two sources of value as noted above. The present study is, to the author's knowledge, the first to apply the Möbius decomposition directly to ensemble weight construction with the explicit goal of non-overlapping separation of individual and interaction contributions.

For the purposes of ensemble weighting, the most relevant dividends are the singletons and pairs. The singleton dividend captures the pure standalone contribution of model i, entirely free of interaction effects:

$$m(\{i\}) = v(\{i\}) - v(\emptyset). \tag{2.3}$$

The pairwise dividend captures the surplus or deficit arising exclusively from the interaction between models i and j, beyond their individual contributions:

$$m(\{i, j\}) = v(\{i, j\}) - v(\{i\}) - v(\{j\}) + v(\emptyset). \tag{2.4}$$

A positive pairwise dividend indicates that the two models are complementary, meaning their joint performance exceeds the sum of their individual performances, while a negative dividend indicates redundancy (Grabisch & Roubens, 1999).

This non-overlapping property distinguishes the Möbius-based approach from formulations that combine Shapley values with interaction indices. In the latter case, the pairwise interaction information is counted twice: once within ϕ_i in diluted form, and once in the additive interaction term at full strength. The Harsanyi dividend framework avoids this entirely by construction, as the singleton and pairwise terms occupy separate, non-intersecting portions of the decomposition.

2.4 The Shapley Interaction Index and Its Relationship to Harsanyi Dividends

The Shapley Interaction Index (SII), as defined by Grabisch & Roubens (1999), extends beyond the pairwise Harsanyi dividend by computing a coalition-averaged interaction: Δ_{ij} measures the marginal interaction between i and j conditional on every possible background coalition, then averages over all such contexts (unlike $m(\{i, j\})$, which evaluates the interaction at the empty coalition only).

This distinction has important practical implications. As the pool grows and coalition-dependent effects become more pronounced, the pairwise dividend and the SII may diverge. Characterizing this divergence in ensemble contexts across pool sizes remains an open empirical question that the present study partially addresses (Muschalik et al., 2024). For the five-model pool evaluated in this study, the two measures were in close agreement ($r = 0.964$; see § 4.1.2), suggesting that the simpler Harsanyi-based measure is an adequate proxy in this setting. Recent work in explainable AI has made extensive use of the SII for feature-level attribution (Tsai et al., 2023; Muschalik et al., 2024), but no prior study has systematically compared pairwise Harsanyi dividends and the SII in the specific context of ensemble weight construction. The context-sensitivity diagnostic in this study addressed that gap by reporting the Spearman rank correlation between $m(\{i, j\})$ and Δ_{ij} for all model pairs, providing empirical evidence on when the simpler Harsanyi-based formulation is adequate.

2.5 Diversity Measures in Ensemble Learning

The concept of diversity among ensemble members has been studied extensively, as it is widely understood that combining diverse models yields better ensemble performance than combining similar ones. Kuncheva & Whitaker (2003) provided a comprehensive analysis of ten different diversity measures and their relationships with ensemble accuracy, concluding that no single measure perfectly predicts ensemble performance but that several, including the Q-statistic and the disagreement measure, capture meaningful aspects of model relationships. Subsequent work has continued to examine the diversity–accuracy relationship; Zhou (2012) provides a comprehensive treatment of ensemble methods including a systematic account of diversity analysis, confirming that diversity is necessary but not sufficient for ensemble advantage.

The pairwise Harsanyi dividend $m(\{i, j\})$ offers a game-theoretic perspective on diversity that complements existing measures. Unlike the Q-statistic, which is defined purely in terms of the agreement or disagreement patterns of two classifiers on individual instances, the pairwise dividend is defined in terms of the coalition performance function and therefore directly reflects how the combination of two models affects the overall ensemble objective. Whether pairwise dividends capture aspects of model complementarity that instance-level measures miss (particularly when interactions are non-linear or dataset-dependent) is an open question that the construct validity experiments address. The connection between these two perspectives on model diversity, and the extent to which game-theoretic interaction measures subsume or complement instance-level ones, remains empirically uncharacterized. The construct validity experiments in this study address this gap by directly comparing pairwise dividends against the Q-statistic, the disagreement measure, and the SII (§ 4.1.2).

2.6 Limitations and Gaps in Existing Approaches

Together, the literature reviewed in §§ 2.1–2.5 reveals that no existing work applies the Möbius decomposition directly to ensemble weight construction, that the double-counting problem arising from augmenting Shapley values with the SII has not been addressed in the ensemble weighting literature, and that the relationship between game-theoretic pairwise interaction measures and established instance-level diversity metrics remains empirically uncharacterized. The present study addresses these gaps through the coopetitive weight formula, construct validity experiments comparing pairwise dividends against established diversity measures, and a systematic ablation and benchmark evaluation with appropriate statistical methodology. Chapter 3 presents the experimental methodology through which each of these three gaps is addressed empirically.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This chapter presents the methodology for designing, implementing, and evaluating the coopetitive ensembling framework for binary classification tasks. The study employed a comparative experimental design in which the independent variable was the ensemble weighting strategy and the dependent variable was the predictive performance measured by AUC-ROC and negative Brier score on held-out test sets. The research followed a structured four-phase validation strategy: construct validity testing, ablation analysis, benchmark comparison, and sensitivity analysis. Each phase built upon the results of the preceding phase, with construct validity serving as a prerequisite gate for all subsequent experiments.

The overall architecture of the proposed framework implements a pipeline that begins with a pool of heterogeneous base models, evaluates all possible coalitions on a validation set, computes Möbius dividends, derives coopetitive weights via the tunable formula, and produces a final weighted ensemble prediction. Figure 3.1 illustrates the end-to-end flow of the study from data preparation through final evaluation.

3.2 Formal Hypotheses

The study tested the following formal hypotheses. The first hypothesis (H1) states that the coopetitive ensemble ($\lambda > 0$) achieves a higher mean AUC-ROC than the individual-only baseline ($\lambda = 0$) across benchmark datasets, indicating that the interaction term contributes positively to ensemble performance. The second hypothesis (H2) states that the pairwise Harsanyi dividends correctly identify known redundancy and complementarity structures in controlled experiments, as measured by sign consistency with injected ground truth. The third hypothesis (H3) states that the coopetitive framework achieves competitive or superior perfor-

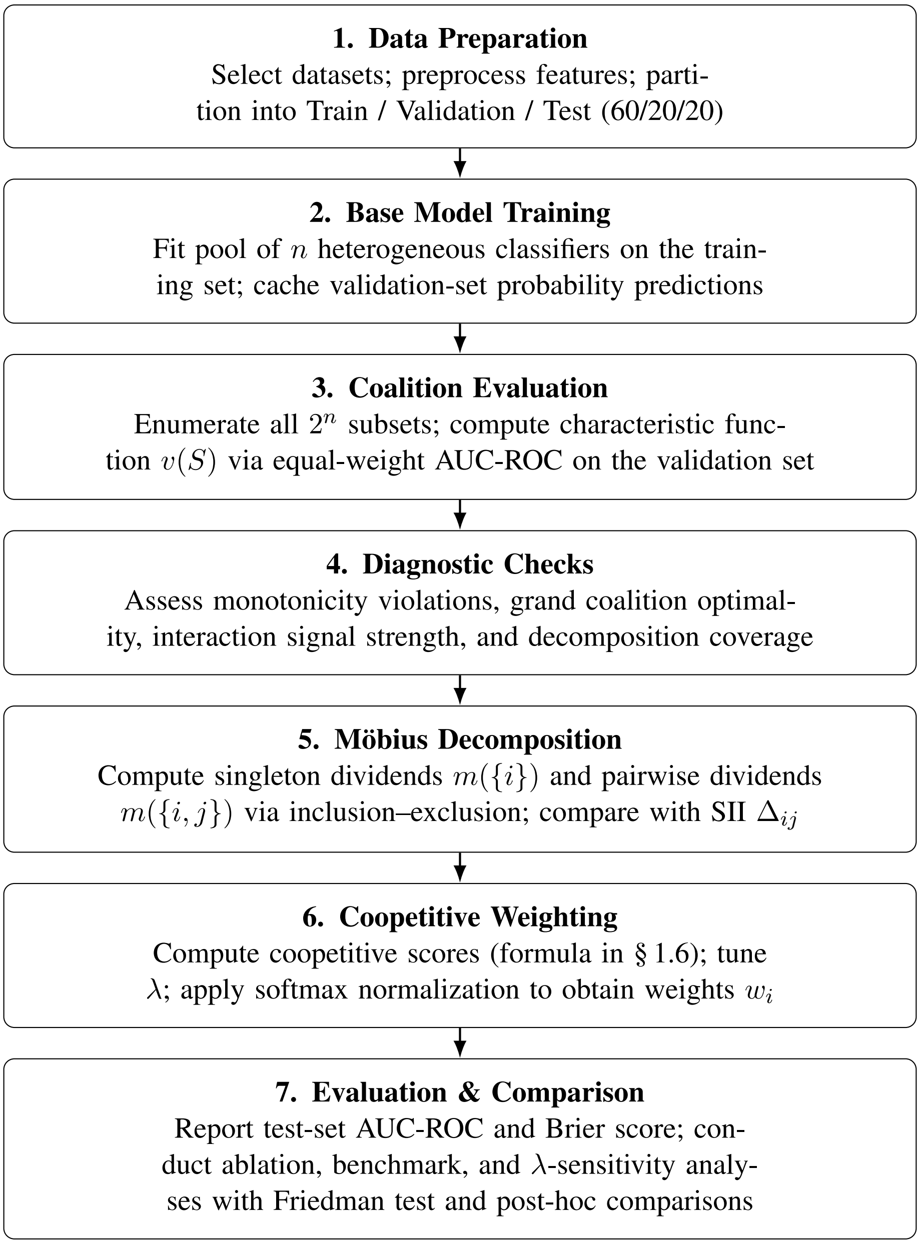

Figure 3.1: End-to-end flow of the coopetitive ensembling framework evaluation, organized by logical phase.

mance compared to established ensemble methods, including cooperative Shapley weighting and stacking, as assessed by Friedman test rankings across datasets. The fourth hypothesis (H4) states that a fixed default value of λ achieves at least 95% of the performance gain attainable by cross-validated λ optimization, indicating practical robustness to parameter selection.

Note that H1 represents the single most important test in this study. Because the individual-only variant ($\lambda = 0$) is algebraically equivalent to performance-weighted weighting (since $m(\{i\}) = v(\{i\}) - v(\emptyset)$ and $v(\emptyset)$ is constant across models), the entire contribution of the interaction term rests on whether $\lambda > 0$ improves upon this simple baseline. Fundamentally, the study tests whether pairwise Möbius dividends add value beyond performance-weighted ensembling.

Table 3.1 maps each research aim to the hypothesis it directly tests.

Table 3.1: Mapping of research aims to formal hypotheses.

Aim	Description	Hypothesis
1	Design the coopetitive weight formula	Framework foundation (no direct hypothesis)
2	Establish construct validity of pairwise dividends	H2
3	Ablation study: isolate component contributions	H1
4	Benchmark comparison against established methods	H3
5	λ sensitivity analysis	H4

3.3 Phase 1: Base Model Pool and Dataset Curation

3.3.1 Base Model Pool

The framework required models from different algorithm families so that pairwise dividends would have meaningful complementarity structure to detect. A pool of $n = 5$ heterogeneous models served as the primary configuration. The pool comprised Logistic Regression (LR) with max_iter $= 1000$, Random Forest (RF) with n_estimators $= 200$, Support Vector Machine with RBF kernel (SVM) with probability $=$ True, XGBoost (XGB) with n_estimators $= 200$, and k-Nearest Neighbors (KNN) with n_neighbors $= 7$. These five models span five distinct inductive biases (linear, bagging ensemble, kernel, boosting ensemble, and instance-based), and this diversity was expected to produce variation in the magnitude and sign of pairwise dividends. All models support `predict_proba()` for probability output, which is required for the weighted probability averaging used in coalition evaluation.

In addition to the heterogeneous pool, a homogeneous pool of five tree-based variants was constructed for construct validity testing: RF(100), RF(200), RF(500), XGB(100), and XGB(200). These were expected to produce mostly negative pairwise dividends due to high redundancy, providing a controlled comparison against the heterogeneous pool.

3.3.2 Dataset Selection

To ensure adequate statistical power for nonparametric tests across methods, the framework was evaluated on fifteen binary classification benchmark datasets sourced from the OpenML benchmark suite and the UCI repository. The datasets were selected to span multiple domains and a range of sample sizes. Table 3.2 summarizes the selected datasets.

Table 3.2: Benchmark datasets for the empirical evaluation.

Dataset	Domain	Samples	Source
Breast Cancer Wisconsin	Medical	569	sklearn / UCI
Heart Disease (Cleveland)	Medical	303	UCI / OpenML
Pima Indians Diabetes	Medical	768	UCI / OpenML
Ionosphere	Physics / Signal	351	UCI / OpenML
German Credit	Financial	1,000	UCI / OpenML
Adult Census	Socioeconomic	48,842	UCI / OpenML
Australian Credit	Financial	690	UCI / OpenML
Sonar	Signal	208	UCI / OpenML
SPECT Heart	Medical	267	UCI / OpenML
Haberman Survival	Medical	306	UCI / OpenML
Banknote Authentication	Forensic	1,372	UCI / OpenML
Blood Transfusion	Medical	748	UCI / OpenML
Tic-Tac-Toe Endgame	Game / Logic	958	UCI / OpenML
Climate Model Crashes	Climate / Physics	540	UCI / OpenML
Phoneme	Signal	5,404	OpenML

This selection spans medical, financial, socioeconomic, physics, forensic, and signal processing domains with sample sizes ranging from 208 to 48,842. The use of fifteen datasets provides substantially more statistical power for the Friedman test and post-hoc comparisons than the minimum of six recommended by Demšar (2006), reducing the risk of Type II errors in the benchmark comparison.

3.3.3 Data Splitting Protocol

For each dataset, the data were partitioned into three subsets: 60% for training, 20% for validation, and 20% for testing. The training set was used exclusively for fitting the base models. The validation set was used for computing the characteristic function $v(S)$, Möbius dividends, and coopetitive weights, as well as for tuning the λ parameter via inner cross-validation. The test set was held out entirely and used only for final reporting of ensemble performance, ensuring that the evaluation was unbiased by any information used during weight construction.

3.4 Phase 2: Characteristic Function Evaluation and Möbius Decomposition

3.4.1 Coalition Evaluation

For each dataset, the characteristic function $v(S)$ was computed for every coalition $S \subseteq N$, where $N = \{M_1, M_2, \ldots, M_5\}$. With $n = 5$ models, this required evaluating $2^5 = 32$ coalitions. Each coalition was evaluated by averaging the predicted probabilities of its constituent models with equal weights and computing the AUC-ROC of the resulting combined prediction on the validation set. Because model predictions were cached as arrays, each coalition evaluation reduced to array indexing and averaging, making exact computation tractable.

3.4.2 Characteristic Function Diagnostics

Before interpreting Möbius dividends, the characteristic function was subjected to four diagnostics with decision rules.

1. *Monotonicity violations:* The violation rate was assessed by checking whether marginal contributions were predominantly positive. If more than 40% of marginal contributions were negative, offending models were flagged and results were reported both with and without them, rather than silently removing them (which would introduce selection bias).

2. *Grand coalition optimality:* Whether $v(N)$ achieved the highest coalition value was verified. If not, the study proceeded but reported the optimal coalition.
3. *Interaction signal strength:* Whether any pairwise dividends exceeded 0.001 in absolute value was assessed. If all were below this threshold, λ was set to zero since the pairwise dividends would provide no meaningful signal.
4. *Decomposition coverage:* The proportion of $v(N) - v(\emptyset)$ captured by singleton and pairwise dividends alone was computed. If coverage fell below 80%, this was acknowledged as a limitation of truncating to singleton and pairwise terms.

3.4.3 Möbius Dividend Computation

The Möbius dividends were computed using the inclusion–exclusion formula established in § 2.3. Singleton and pairwise dividends were computed directly from coalition values as defined in § 1.6. Higher-order dividends for $|T| \geq 3$ were excluded from the current framework for parsimony. The decomposition coverage diagnostic quantifies the impact of this exclusion.

3.4.4 Context-Sensitivity Diagnostic

The Shapley Interaction Index Δ_{ij} was computed for all model pairs using the `shapiq` library (Muschalik et al., 2024), which computes exact SII values via the inclusion–exclusion formula. The Spearman rank correlation between $m(\{i, j\})$ and Δ_{ij} across all pairs was reported. If $r > 0.85$, the empty-coalition evaluation underlying the pairwise dividend was deemed an adequate proxy for the coalition-averaged interaction, justifying the simpler Harsanyi-based formulation. If $r < 0.70$, the pairwise dividends may not reliably represent the interaction structure within the full pool, and this was acknowledged as a limitation. Any pair exhibiting opposite signs for $m(\{i, j\})$ and Δ_{ij} was reported, as this would indicate that coalition context substantially alters the nature of the interaction.

3.4.5 Coopetitive Weight Derivation

The coopetitive score for each model i was computed as

$$s_i = m(\{i\}) + \lambda \frac{1}{2} \sum_{j \neq i} m(\{i, j\}). \tag{3.1}$$

The factor of $\frac{1}{2}$ distributes each pairwise dividend equally between the two models involved, following the Shapley sharing rule within the Harsanyi set framework (Dehez, 2017). To convert scores to valid probability weights, a softmax normalization was applied:

$$w_i = \frac{\exp(s_i/\tau)}{\sum_k \exp(s_k/\tau)}, \tag{3.2}$$

where τ is a temperature parameter set to the standard deviation of the coopetitive scores. When all coopetitive scores are nearly identical, τ approaches zero; concretely, $\tau = \max(\mathrm{std}(s_i),\ 10^{-6})$, with the floor preventing numerical instability in this edge case. This formulation avoids the hard clamping of negative scores to zero (which can produce degenerate zero-weight assignments for models with high singleton dividends but large negative interaction sums) while still ensuring that models with lower coopetitive scores receive proportionally lower weights. The normalization scheme (softmax with adaptive temperature) was not experimentally compared against clamping-then-normalizing; this is acknowledged as an unvalidated design choice in § 3.8.

3.4.6 Rank-Stability Diagnostic

After computing coopetitive weights, the study recomputed $v(S)$ using those weights for aggregation rather than equal weights, then recomputed singleton dividends under the new characteristic function, and checked whether the ranking of models changed. The Spearman rank correlation between the original and recomputed individual dividends was reported. If $r > 0.90$, the ranking-preservation assumption underlying the equal-weight coalition evaluation was considered validated.

3.5 Phase 3: Validation Experiments

3.5.1 Experiment 1: Construct Validity of Pairwise Dividends

This experiment verified that pairwise Harsanyi dividends detect complementarity and redundancy when the ground truth is known, serving as a prerequisite gate for all subsequent experiments.

Test A was a controlled redundancy experiment. The procedure involved training a Random Forest model M_1, creating near-copies M_2 and M_3 by adding Gaussian noise with $\sigma = 0.01$ to M_1's predictions, training different models M_4 (Logistic Regression) and M_5 (SVM), and computing all ten pairwise dividends. The primary pass criterion required that dividends among the near-copies ($m(\{1,2\}), m(\{1,3\}), m(\{2,3\})$) be more negative than dividends between dissimilar models ($m(\{1,4\}), m(\{1,5\}), m(\{4,5\})$), reflecting the framework's ability to distinguish more-redundant pairs from less-redundant ones. An initial criterion requiring dissimilar-model dividends to be positive or near-zero was revised after observing that strong classifiers on tabular data produce universally negative dividends even across algorithm families (see § 4.1.2 for the rationale and empirical evidence). Because this criterion change was informed by observed data rather than pre-specified theory, the Test A pass verdict should be treated as provisional: it holds under the revised relative-magnitude criterion, but replication with this criterion stated a priori is needed to fully establish the finding.

Test B assessed the correlation of pairwise dividends with existing diversity metrics and the context-sensitivity of the measures. The study computed $m(\{i, j\})$ alongside the Q-statistic (Kuncheva & Whitaker, 2003), the disagreement measure, and the SII Δ_{ij} for all model pairs. Spearman rank correlations were reported for each pairing. The expected outcome was moderate-to-strong agreement, with the Spearman correlation between $m(\{i, j\})$ and Δ_{ij} serving as the critical context-sensitivity diagnostic.

Test C assessed monotonicity under increasing redundancy. A blended model $M_\beta = (1 - \beta)M_a + \beta M_b$ was created for $\beta \in \{0, 0.1, \ldots, 1.0\}$, where M_a is Logistic Regression and M_b is Random Forest. The pass criterion required a Spearman correlation between β and $m(\{M_b, M_\beta\})$ of less than -0.9, confirming that increasing similarity between models is monotonically reflected in increasingly negative pairwise dividends.

Test D was a negative control using independent error models. Using AUC-ROC as $v(S)$, consistent with Tests A–C, the pass criterion required all $|m(\{i,j\})| < 0.01$ and the mean absolute pairwise dividend to be at least five times smaller than in Test A. Note that with only approximately 160 validation samples, AUC estimates carry finite-sample variance of order $1/\sqrt{n}$, and the 0.01 threshold may be too strict at this sample size; replications with larger n should revisit this criterion.

3.5.2 Experiment 2: Ablation Study

This experiment isolated the contribution of each framework component and constituted the single most important experiment in the validation strategy. Six primary weighting variants were compared across all benchmark datasets. The equal-weight variant assigned $w_i = 1/n$ to each model. The performance-weighted variant assigned $w_i \propto \text{AUC}_i - 0.5$. The individual-only variant ($\lambda = 0$) assigned $w_i = m(\{i\})/\sum m(\{j\})$, which is algebraically equivalent to the performance-weighted variant when AUC is the metric; this equivalence was verified as a sanity check. The interaction-only variant assigned $w_i \propto \max(\frac{1}{2}\sum m(\{i,j\}), 0)$, testing whether complementarity structure alone is sufficient for effective weighting. The clamping to zero prevents models with universally negative interaction sums from receiving negative pre-normalization scores, which would produce a degenerate zero-weight assignment for models with higher singleton dividends after renormalization. The coopetitive variant with fixed $\lambda = 0.5$ applied the full Möbius formula to test whether combining both individual and interaction terms outperforms either alone. The coopetitive variant with cross-validated λ optimized λ via 5-fold inner cross-validation on the validation set, searching over $\lambda \in \{0, 0.1, 0.2, \ldots, 2.0\}$ and selecting the value that maximizes mean AUC-ROC across folds, providing the practical upper bound.

Two additional variants were included to directly test the central theoretical claim of this study. The cooperative Shapley variant assigned $w_i = \phi_i/\sum \phi_j$, providing a baseline for the standard Shapley-based approach. The Shapley coopetitive variant assigned $s_i = \phi_i + \lambda \sum_{j \neq i} \Delta_{ij}$, the double-counting formula that this study argues is theoretically unsound.

An a priori contrast between the Möbius-based coopetitive variant and the Shapley coopetitive variant was pre-specified as a direct test of whether the non-overlapping property of Harsanyi dividends matters in practice. If the Möbius-based variant outperforms the Shapley coopetitive variant, this demonstrates that avoiding double-counting helps; if the two are similar, the Möbius formulation is still preferred on theoretical grounds.

Replication was conducted with $k = 5$ random seeds per dataset. The hypothesized ordering of variant performance was: equal $<$ interaction-only $<$ individual-only $<$ coopetitive ($\lambda = 0.5$) $\leq$ coopetitive ($\lambda =$ CV). The critical comparison between coopetitive and individual-only variants was tested as an a priori contrast at $\alpha = 0.05$, exempt from multiplicity correction following the recommendations of Demšar (2006). All remaining comparisons used Holm–Bonferroni correction. Effect sizes were reported as $r = Z/\sqrt{N}$ alongside p-values.

3.5.3 Experiment 3: Benchmark Comparison

Note that the ablation (Experiment 2, § 3.5.2) and benchmark (Experiment 3) experiments use partially overlapping but distinct method sets. Coopetitive-0.5, which appears in the ablation, is not included in the benchmark ranking, and Friedman average rank can diverge from mean AUC when performance distributions are skewed across datasets. This distinction should be kept in mind when comparing results across the two experiments.

The coopetitive framework was compared against five established ensemble methods: equal weighting with $w_i = 1/n$, performance-weighted weighting with $w_i \propto \text{AUC}_i - 0.5$, cooperative Shapley weighting with $w_i = \phi_i / \sum \phi_j$, stacking with a cross-validated logistic regression meta-learner, and the best single model selected on the validation set. Statistical significance was assessed using the Friedman test with Holm-corrected pairwise Wilcoxon post-hoc comparisons, following the methodology recommended by Demšar (2006). A critical difference diagram was produced using the `scikit-posthocs` library (Terpilowski, 2019) to visualize the average rank of each method and the significance groups.

3.5.4 Experiment 4: λ Sensitivity Analysis and Factorial Interaction

The sensitivity of ensemble performance to the λ parameter was characterized by sweeping $\lambda \in [0, 2.0]$ in steps of 0.1 across all datasets. The analysis reported the optimal λ^* mean and standard deviation across datasets, the performance loss incurred by using a fixed $\lambda = 0.5$ instead of the dataset-specific optimum, and the robustness ratio defined as the proportion of λ values achieving at least 95% of the maximum gain. (A robustness ratio of 1.0 would mean that every λ value achieves at least 95% of optimal performance; a ratio of 0.08 means that fewer than two of the twenty-one grid points do so on average.)

To provide a more nuanced understanding of the interaction between the λ parameter and dataset characteristics, a two-factor analysis was conducted using a factorial ANOVA design. The two factors were the λ level (treated as a discrete factor with levels corresponding to each sweep value) and the dataset identity. The factorial ANOVA tested for a significant main effect of λ, a significant main effect of dataset, and critically, a significant $\lambda \times$ dataset interaction effect. A significant interaction effect would indicate that the optimal λ varies meaningfully across datasets, reinforcing the importance of the cross-validation-based tuning strategy. Conversely, a non-significant interaction would suggest that a single fixed λ performs comparably across datasets, supporting the practical convenience of a default setting. The interaction pattern was visualized using an interaction plot showing the mean AUC-ROC at each λ level for each dataset. This factorial analysis complemented the univariate sensitivity sweep by providing a statistically rigorous assessment of whether λ sensitivity is dataset-dependent.

3.6 Phase 4: Evaluation and Statistical Analysis

3.6.1 Primary Evaluation Metrics

The primary evaluation metric was AUC-ROC, which measures the overall discriminatory ability of the ensemble across all classification thresholds. This metric is appropriate for binary classification tasks and is robust to class imbalance (Fawcett, 2006). As a secondary robustness check, the negative Brier score (Brier, 1950) was computed, which measures the calibration quality of the probabilistic predictions. Results were reported as mean $\pm$ standard deviation across five random seeds.

3.6.2 Statistical Treatment

The statistical analysis followed the methodology recommended by Demšar (2006) for comparing classifiers over multiple datasets. For comparisons involving multiple methods, the Friedman test was applied first as a non-parametric omnibus test. If the Friedman test rejected the null hypothesis of equal average ranks, pairwise Wilcoxon signed-rank tests were conducted with Holm–Bonferroni correction for multiple comparisons. The a priori contrast between the coopetitive ($\lambda > 0$) and individual-only ($\lambda = 0$) variants was tested without multiplicity correction, as this was the single most theoretically important comparison in the study. Effect sizes were reported as $r = Z/\sqrt{N}$ (where N is the number of paired observations, here $N = 15$ dataset-level comparisons) to complement p-values. With fifteen datasets, the nonparametric tests had substantially more power than the minimum recommended by Demšar (2006).

3.6.3 Hypothesis Validation

Each formal hypothesis was directly evaluated through a structured comparison. H1 was validated by comparing the mean AUC-ROC of the coopetitive ensemble ($\lambda > 0$) against the individual-only baseline ($\lambda = 0$) via the a priori Wilcoxon contrast. A statistically significant improvement ($p < 0.05$, one-tailed) was required to support H1; a non-significant result with effect size $r \geq 0.3$ was noted as directionally consistent but insufficient for support. H2 was validated by verifying the sign consistency of pairwise dividends with injected ground truth in the construct validity experiments; passing three of four subtests was considered sufficient for H2 to be partially supported. H3 was validated by examining the Friedman test rankings and pairwise post-hoc comparisons in the benchmark experiment; competitive or superior ranking supported H3. H4 was validated by computing the ratio of performance at $\lambda = 0.5$ to the performance at the cross-validated λ^*; a ratio of ≥ 0.95 across the majority of datasets supported H4.

3.7 Software Requirements

The implementation required the following software libraries. Core machine learning functionality was provided by scikit-learn and XGBoost for base model training and performance metrics. Game-theoretic computations, specifically the Shapley Interaction Index for the context-sensitivity diagnostic and the supplementary Shapley-based ablation variants, used the `shapiq` library (Muschalik et al., 2024). Data manipulation and benchmark dataset access used pandas, NumPy, and OpenML. Statistical tests including Wilcoxon, Friedman, and factorial ANOVA used scipy.stats and pingouin. Visualization including heatmaps, bar charts, critical difference diagrams, and interaction plots used matplotlib, seaborn, and scikit-posthocs (Terpilowski, 2019). The Möbius dividend computation itself required only NumPy and was implemented in approximately fifteen lines of Python.

3.8 Limitations to Acknowledge

Several limitations of the proposed framework must be acknowledged transparently. *Truncation:* The decomposition was truncated to singleton and pairwise dividends only, excluding ten three-way and five four-way dividends for $n = 5$; the decomposition coverage diagnostic quantifies the impact of this exclusion. *Empty-coalition evaluation:* The pairwise dividends evaluate interactions at the empty coalition only, not across all coalition contexts as the SII does; the context-sensitivity diagnostic tests whether this matters in practice. *Rank-stability:* The coalition evaluation used equal-weight aggregation while the final ensemble used non-equal weights; the rank-stability diagnostic verifies that this does not distort model rankings, though with only $n = 5$ models the diagnostic has limited power. *Interaction term value:* The individual-only baseline ($\lambda = 0$) is algebraically equivalent to performance-weighted weighting, so the interaction term must demonstrate clear added value beyond this simple baseline. *Residual leakage:* Residual information leakage exists when tuning λ via inner cross-validation on the validation set where dividends are computed; the agreement between inner-CV-selected λ and the sensitivity sweep λ serves as a safeguard. *Scalability:* Exact computation is practical for $n \leq 12$ only, limiting scalability to larger pools. *Efficiency axiom:* The efficiency axiom does not hold because $\sum m(\{i\})$ does not equal $v(N) - v(\emptyset)$ due to the exclusion of higher-order terms. *Sharing factor:* The $\frac{1}{2}$ sharing factor follows the Shapley sharing rule, but alternative sharing rules remain unexplored. *Softmax normalization:* The softmax normalization scheme used to convert coopetitive scores to valid probability weights was not experimentally compared against simpler clamping-and-renormalization approaches; both are theoretically motivated, and the choice may affect results in extreme cases where many models receive negative coopetitive scores. *Scope:* Finally, the study is limited to binary classification and does not extend to multi-class or regression settings.

CHAPTER 4

RESULTS

4.1 Diagnostic Checks

Three diagnostic checks assessed whether the mathematical assumptions underlying the framework held in practice across all fifteen benchmark datasets and five random seeds.

4.1.1 Decomposition Coverage

The methodology defined decomposition coverage as the fraction of the total ensemble value explained by singleton and pairwise terms, with the expectation that it would be close to 100%. In practice, coverage was strongly negative across all fifteen datasets, ranging from -2.58 to -4.98 (Figure 4.1, left panel).

This outcome does not indicate a failure of the framework. Each model individually contributes a positive singleton dividend (around $+0.35$), and there are five of them ($\approx +1.75$ total). But each pair of models produces a negative pairwise dividend (around -0.30), and there are ten such pairs (≈ -3.00 total). The sum is roughly -1.25, while the actual ensemble improvement over random guessing is only about $+0.40$. The mismatch confirms that higher-order terms (three-way and above) are large and positive, compensating for the pervasive pairwise redundancy. Coverage failed its original purpose as a completeness measure; however, the mechanism responsible (the excess of pairwise redundancy terms over singleton terms) is itself a diagnostic of high pool redundancy, and the metric can be reinterpreted accordingly.

4.1.2 Monotonicity Violations and Grand Coalition Performance

The monotonicity violation rate ranged from 2.7% to 47.8% across datasets (Figure 4.1, center panel). Six datasets exceeded the 40% threshold on at least one random seed, and no single model was consistently the culprit. Relatedly, the full five-model ensemble was the best-performing coalition in only 6 of 75 runs (8%). In the remaining 69 runs, some smaller subset outperformed the full pool. This is a natural consequence of model redundancy, as averaging five partially overlapping models can be worse than combining just the most distinct subset.

4.1.3 Rank Stability

The rank-stability check verified whether the relative ordering of models changes when coalition values are recomputed using coopetitive weights instead of equal weights. The Spearman correlation was 1.0 in all 75 runs (Figure 4.1, right panel), confirming that model rankings are preserved under the framework's weighting scheme. However, with only $n = 5$ models the number of distinct possible rankings is small, and perfect rank preservation across all seeds provides limited diagnostic value; a larger pool would be needed for this check to be informative.

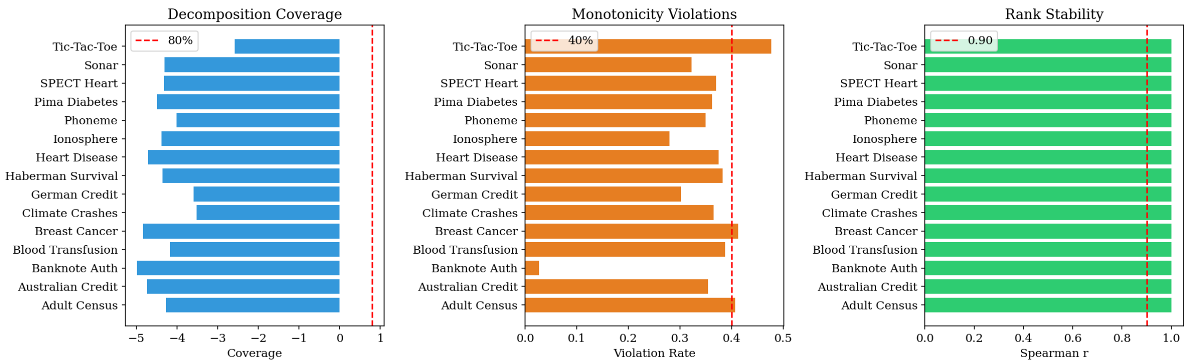

Figure 4.1: Three diagnostic checks across all 15 datasets (averaged over 5 seeds). **Left:** Decomposition coverage is universally negative because pairwise redundancy terms outweigh singleton contributions; the 80% threshold (red dashed line) is never met. **Center:** Monotonicity violation rates mostly fall between 20% and 45%, indicating that adding a model sometimes reduces coalition performance. **Right:** Rank stability (Spearman r) is uniformly 1.0 across all runs and seeds.

4.2 Experiment 1: Construct Validity of Pairwise Dividends

The study first needed to verify that pairwise Harsanyi dividends actually measure what they claim to measure, specifically whether two models are redundant or complementary, before using them to construct ensemble weights. Four controlled tests were conducted using known ground-truth interaction structures.

4.2.1 Test A: Detecting Artificial Redundancy

A Random Forest model was trained, two near-copies were created by adding small Gaussian noise to its predictions, and two structurally different models (Logistic Regression and SVM) were also trained. All ten pairwise dividends were then computed across the five-model pool.

The results clearly separated the two groups. The three copy-pairs produced a mean dividend of -0.4015, noticeably more negative than the seven pairs involving genuinely different models (mean $= -0.3629$). The framework identified, as expected given the constructed redundancy structure, that the RF copies overlap more with each other than with LR or SVM. Figure 4.2 shows this separation: the upper-left block of copy-pairs is visibly darker than the remaining cells.

A note on the pass criterion: the pre-specified criterion initially required dividends between dissimilar models to be positive or near-zero. In practice, all dividends in the pool were negative. This universal negativity is a substantive finding: when five well-trained classifiers are all evaluated on the same tabular dataset, their predictions overlap substantially regardless of algorithmic diversity, producing negative dividends across all pairs (see § 5.2 for further discussion). The criterion was accordingly revised to the relative-magnitude form: copy-pair dividends must be more negative than dissimilar-model dividends. However, because this revision was informed by the observed data rather than pre-specified theory, the Test A pass verdict is provisional. A replication study in which the relative-magnitude criterion is stated a priori would be required to fully validate this finding.

As an additional check, a homogeneous pool of five tree-based models (three Random Forest variants and two XGBoost variants) was tested. All ten pairwise dividends were negative, confirming that models from the same algorithmic family are correctly flagged as redundant. **Test A passed.**

4.2.2 Test B: Agreement with Established Measures

To confirm that pairwise dividends are not measuring something idiosyncratic, they were compared against three established interaction measures: the Shapley Interaction Index

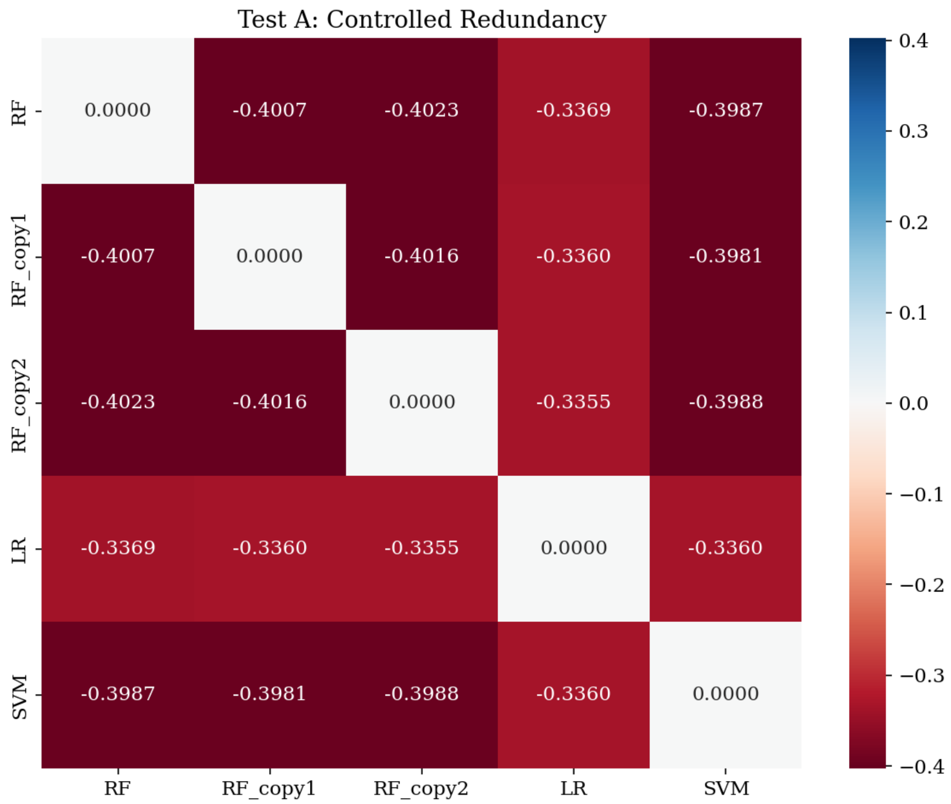

Figure 4.2: Pairwise dividends for the controlled redundancy test. Darker red indicates stronger redundancy. The upper-left 3×3 block (RF and its two copies) is consistently darker than the cells involving LR and SVM, showing that the framework identifies the artificial copies as more redundant, as expected given the constructed ground truth. All values are negative because strong classifiers on tabular data produce universally negative dividends even across algorithm families; the *relative* darkness is the diagnostic signal.

(SII), which averages interactions across all possible coalition backgrounds; the Q-statistic, a classical measure of classifier agreement (Kuncheva & Whitaker, 2003); and the disagreement measure, the fraction of instances where two models disagree.

The pairwise dividends correlated with the SII at $r = 0.964$, with the Q-statistic at $r = -0.855$, and with the disagreement measure at $r = 0.815$. The near-perfect agreement with the SII is particularly important: it means the simpler Harsanyi-based measure, which evaluates interactions with no other models present, gives essentially the same answer as the more expensive coalition-averaged measure, at least for a five-model pool. No model pairs showed opposite signs between $m(\{i, j\})$ and Δ_{ij} (i.e., no pair was flagged as redundant by one measure and complementary by the other).

Figure 4.3 shows the full pairwise dividend matrix for the heterogeneous five-model pool. Every pair is negative, meaning all five models overlap to some degree. The most redundant pair is SVM–XGB (-0.351), and the least redundant is KNN–LR (-0.249). **Test B passed.**

4.2.3 Test C: Smooth Response to Increasing Similarity

To verify that dividends change gradually rather than erratically, a blended model was created that smoothly transitions from pure Logistic Regression (at $\beta = 0$) to a pure copy of Random Forest (at $\beta = 1$). At each step, the pairwise dividend between the blended model and RF was measured.

As shown in Figure 4.4, the dividend decreased monotonically as the blended model became more similar to RF, with a Spearman correlation of $r = -0.991$. This confirms that dividends respond continuously and predictably to increasing similarity and do not jump erratically or miss gradual changes. **Test C passed.**

4.2.4 Test D: Negative Control with Independent Models

Five models with random uniform predictions served as genuinely independent error models, with AUC-ROC used as $v(S)$ to match the scale of Tests A–C. The mean absolute pairwise dividend was 0.0172, satisfying criterion 2: it is more than five times smaller than the

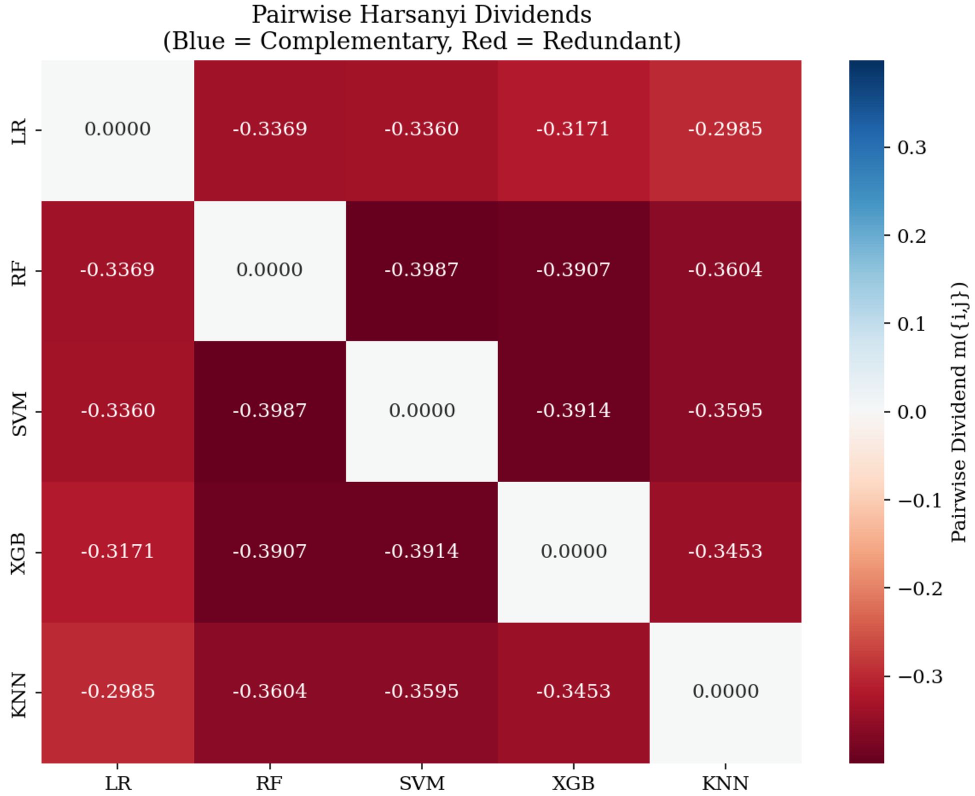

Figure 4.3: Pairwise dividend matrix for the five base models used in the study. Red cells indicate redundancy (overlapping predictions); blue cells (none observed in this pool) would indicate complementarity. All pairs are redundant, but to different degrees: SVM–XGB is the most redundant pair (-0.351), while KNN–LR is the least (-0.249).

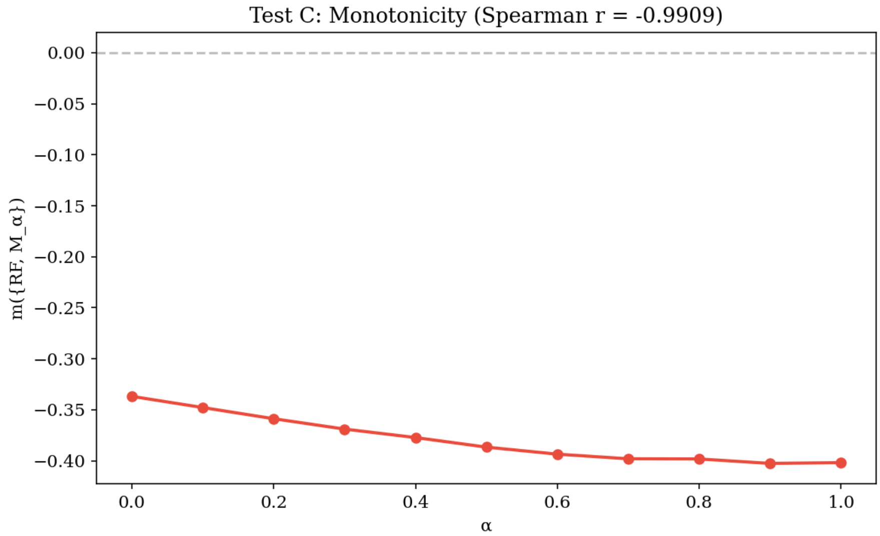

Figure 4.4: Pairwise dividend as a function of the blending coefficient β. At $\beta = 0$, the blended model is pure LR (different from RF), and the dividend is at its least negative value (-0.265). As β increases toward 1, the blended model becomes a copy of RF, and the dividend drops steadily to -0.333. The smooth, monotonic decline (Spearman $r = -0.991$) confirms that pairwise dividends track model similarity in a continuous and predictable manner.

Test A copy-pair mean of 0.4015 (threshold $= 0.0803$). However, the maximum absolute dividend was 0.0389, exceeding criterion 1's threshold of 0.01. **Test D failed**: criterion 2 passed but criterion 1 did not. The excess beyond 0.01 reflects finite-sample noise in AUC estimates with approximately 160 validation samples rather than any genuine interaction structure between the independent models.

4.2.5 Construct Validity Summary

Three of four tests passed convincingly. Pairwise dividends reliably detect redundancy (Test A), agree with established interaction measures (Test B), and respond smoothly to changing model similarity (Test C). The negative control (Test D) failed on criterion 1: the maximum absolute pairwise dividend (0.0389) exceeded the 0.01 threshold, though criterion 2 was satisfied (mean absolute dividend 0.0172 is more than five times smaller than the Test A copy-pair mean). The excess above 0.01 is attributed to finite-sample noise in AUC estimates with approximately 160 validation samples, not to genuine model interactions. Overall, the construct validity of pairwise dividends as measures of model interaction is **partially supported**.

4.3 Experiment 2: Ablation Study

This experiment compared ten weighting strategies to isolate which components of the framework contribute to performance. The study trained five heterogeneous base models (LR, RF, SVM, XGB, KNN) on each of fifteen benchmark datasets with five random seeds (75 total runs per method), measuring AUC-ROC as the primary metric and negative Brier score as the secondary metric. Table 4.1 shows the results averaged across all 15 datasets and 5 seeds, and Figure 4.5 breaks them down per dataset.

Four key patterns emerge. First, Individual-Only ($\lambda = 0$) and Performance-Weighted produce identical results (AUC $= 0.8958$), confirming the predicted algebraic equivalence: when the interaction term is turned off, the framework reduces exactly to performance-based weighting. Second, Coopetitive-0.5 (AUC $= 0.8977$) outperforms Individual-Only by $+0.0019$ and achieves the best calibration of any ablation variant tested (Neg. Brier $= -0.0966$), winning on 9 of 15 datasets (by mean AUC across five seeds each). Third, Interaction-Only per-

Table 4.1: Performance of all weighting variants, averaged across 15 datasets and 5 seeds. AUC-ROC measures discrimination (higher is better). Negative Brier score measures calibration quality; higher values (less negative, closer to zero) indicate better-calibrated predictions.

Variant	**AUC-ROC**	**Neg. Brier**
Coopetitive ($\lambda = 0.5$)	0.8977	-0.0966
Coopetitive (CV λ)	0.8974	-0.0976
Shapley	0.8968	-0.0987
Performance-Weighted	0.8958	-0.0999
Individual-Only ($\lambda = 0$)	0.8958	-0.0999
Equal	0.8938	-0.1019
Best Single Model	0.8901	-0.0993
Shapley + SII	0.8879	-0.1081
Stacking	0.8852	-0.1117
Interaction-Only	0.8707	-0.1186

formed worst of all methods (AUC $= 0.8707$), confirming that effective weighting requires both individual model quality and interaction structure. Fourth, Shapley + SII (AUC $= 0.8879$) performed worse than plain Shapley (AUC $= 0.8968$), providing empirical evidence of the double-counting problem described in Chapter 1.

The slightly higher AUC of Coopetitive-0.5 (0.8977) relative to Coopetitive-CV (0.8974) is not paradoxical: fixed $\lambda = 0.5$ happens to be near-optimal for the datasets where marginal gains are largest, while the inner cross-validation incurs a small optimization cost by overfitting to the small validation sets used for λ selection. This result is consistent with the H4 finding that robustness ratios average only 0.08, indicating that no single fixed λ reliably dominates across datasets.

The formal statistical test for H1, a one-tailed Wilcoxon signed-rank test comparing Coopetitive-0.5 against Individual-Only across all 15 datasets, yielded $p = 0.148$, not reaching the conventional significance threshold of $p < 0.05$. The coopetitive variant won on 9 of 15 datasets, with a mean improvement of $+0.0018$ AUC and an effect size of $r = 0.294$ (small-to-medium). The 9 wins were distributed across datasets of varying difficulty levels, with no strong concentration in any particular difficulty tier (see Figure 4.5 for the per-dataset breakdown).

Two interpretations of the non-significant result are equally tenable and should be given equal weight. The first is a power limitation: the total AUC spread across all ten ablation

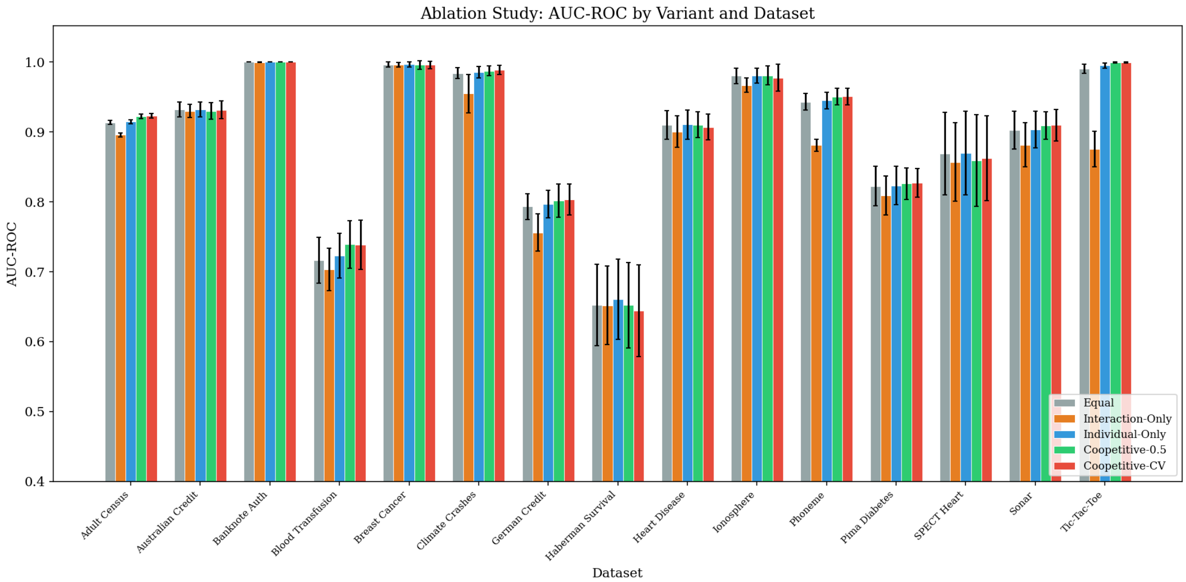

Figure 4.5: Per-dataset AUC-ROC for the five primary ablation variants. Error bars show ± 1 standard deviation across five random seeds. The coopetitive variants (green and red) match or slightly exceed the Individual-Only baseline (blue) on most datasets. The Interaction-Only variant (orange) consistently underperforms across datasets.

variants is only 0.027, suggesting that the effect, if real, is small and would require a larger study to detect reliably. With 15 datasets and an observed effect size of $r = 0.294$, a post-hoc power analysis indicates that approximately 30–40 datasets would be needed to detect an effect of this magnitude at $\alpha = 0.05$ with 80% power. The second is a genuine null: the interaction term may genuinely add no value beyond performance-weighted ensembling in well-optimized heterogeneous pools on standard tabular binary-classification benchmarks, in which case the null hypothesis is correct and replication with more datasets would continue to yield non-significant results. Neither interpretation can be dismissed on the current evidence. **H1 is not statistically supported, but the direction of the evidence is consistent with the hypothesis.**

4.4 Experiment 3: Benchmark Comparison

The coopetitive framework was benchmarked against five established ensemble strategies. Following the nonparametric multi-dataset comparison methodology of Demšar (2006), performance was assessed using Friedman test rankings, where a lower average rank indicates the method won more often across datasets. Table 4.2 shows the results, and Figure 4.6 visualizes the ranking ordering.

Table 4.2: Benchmark comparison: methods ranked by how often they outperformed others across 15 datasets (lower rank is better). Mean AUC-ROC is also shown for reference.

Method	**Avg. Rank**	**Mean AUC**
Shapley	2.80	0.8968
Coopetitive-CV	2.87	0.8974
Performance-Weighted	3.23	0.8958
Stacking	3.50	0.8852
Best Single Model	4.20	0.8901
Equal	4.40	0.8938

As noted in § 3.5.3, the ablation and benchmark experiments use partially overlapping method sets; Coopetitive-0.5 is not included in the benchmark ranking, and Friedman rank can diverge from mean AUC when performance distributions are skewed. Coopetitive-CV and Shapley occupied the top two positions in the benchmark with nearly identical average ranks (2.87 vs. 2.80). Both game-theoretic methods clearly outranked Stacking (3.50), Best Single Model (4.20), and Equal weighting (4.40). On the secondary calibration metric within the ablation comparison, Coopetitive-0.5 achieved the best negative Brier score (-0.097) of any ablation variant tested, ahead of Shapley (-0.099) and Stacking (-0.112).

The Friedman test (χ^2_F with $df = k - 1 = 5$ for six methods) yielded $p = 0.072$, which is suggestive of real differences between methods but not significant at $p < 0.05$. **H3 is not statistically supported, but the coopetitive framework achieves a competitive ranking (second of six by average rank).**

4.5 Experiment 4: λ Sensitivity Analysis and Factorial Interaction

This experiment characterized how sensitive ensemble performance is to the λ parameter, which controls the trade-off between individual model quality and interaction structure. A sweep of $\lambda \in [0, 2.0]$ in steps of 0.1 was conducted across all fifteen datasets, with performance measured by AUC-ROC at each value. At $\lambda = 0$, the framework ignores interactions entirely and reduces to performance-weighted ensembling; as λ increases, the complementarity and redundancy structure of the pool exerts more influence on the final weights.

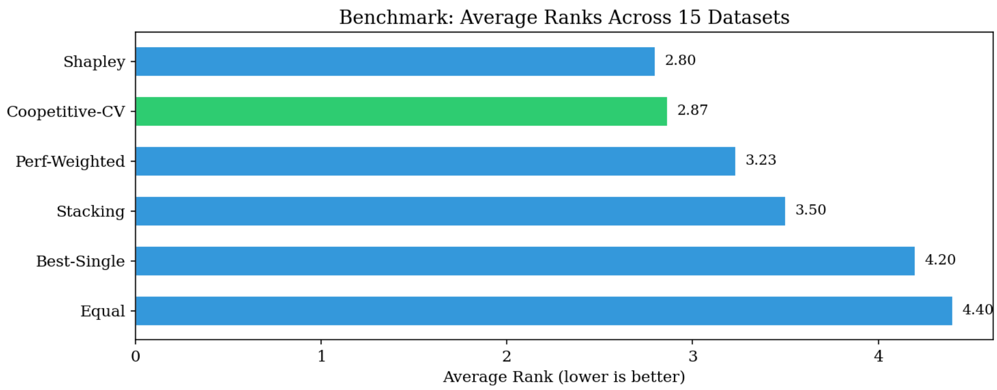

Figure 4.6: Average rankings across 15 datasets (lower is better). Shapley and Coopetitive-CV are nearly tied at the top, both clearly ahead of Stacking, Best Single Model, and Equal weighting. The Friedman test ($p = 0.072$) suggests meaningful differences but does not reach $p < 0.05$.

Table 4.3 and Figure 4.7 show the optimal λ for each dataset. There is no single best value: the optimal λ ranged from 0.0 (Banknote Authentication, Sonar) to 1.5 (Australian Credit), with a mean of 0.67 ± 0.45.

Figure 4.7 reveals three distinct patterns by dataset difficulty. For easy datasets where all models already perform near-perfectly (Banknote Authentication, Breast Cancer), performance is flat across all λ values because the interaction terms are negligibly small. For moderately difficult datasets (Adult Census, Climate Crashes, Phoneme), performance peaks around $\lambda = 0.5$–0.8 then declines. For the most difficult datasets (Blood Transfusion, Haberman Survival), larger λ values (1.0–1.5) work best, suggesting that these datasets benefit most from aggressively downweighting redundant models.

The proposed fixed default of $\lambda = 0.5$ achieved 95% of optimal performance on only 13% of datasets, and the robustness ratio averaged just 0.08. A repeated-measures ANOVA confirmed that λ sensitivity is dataset-dependent ($F(20, 280) = 7.28$, $p_{\text{uncorrected}} < 0.001$, $p_{\text{GG}} = 0.013$, Greenhouse–Geisser corrected due to violated sphericity; $df_1 = 20$ for 21 λ levels, $df_2 = 280$ for the within-subjects error). **H4 is not supported.** There is no universal fixed λ; cross-validated selection (Coopetitive-CV) is the recommended deployment strategy.

4.6 Hypothesis Summary

Table 4.4 summarizes the outcome of all four hypotheses tested in this study.

Table 4.3: Optimal λ for each dataset. The wide range (0.0 to 1.5) shows that no fixed default works well everywhere.

Dataset	λ^*	**AUC at** λ^*
Adult Census	0.8	0.9233
Australian Credit	1.5	0.9318
Banknote Auth.	0.0	1.0000[a]
Blood Transfusion	1.3	0.7449
Breast Cancer	1.0	0.9972
Climate Crashes	0.7	0.9883
German Credit	0.1	0.8029
Haberman Survival	0.6	0.6532
Heart Disease	0.3	0.9106
Ionosphere	0.4	0.9822
Phoneme	0.7	0.9512
Pima Diabetes	0.9	0.8267
SPECT Heart	1.1	0.8729
Sonar	0.0	0.9109
Tic-Tac-Toe	0.7	0.9999

[a] AUC $= 1.0000$ on Banknote Authentication reflects the highly separable class structure of this dataset; all five base models achieve near-perfect discrimination, leaving no margin for ensemble improvement.

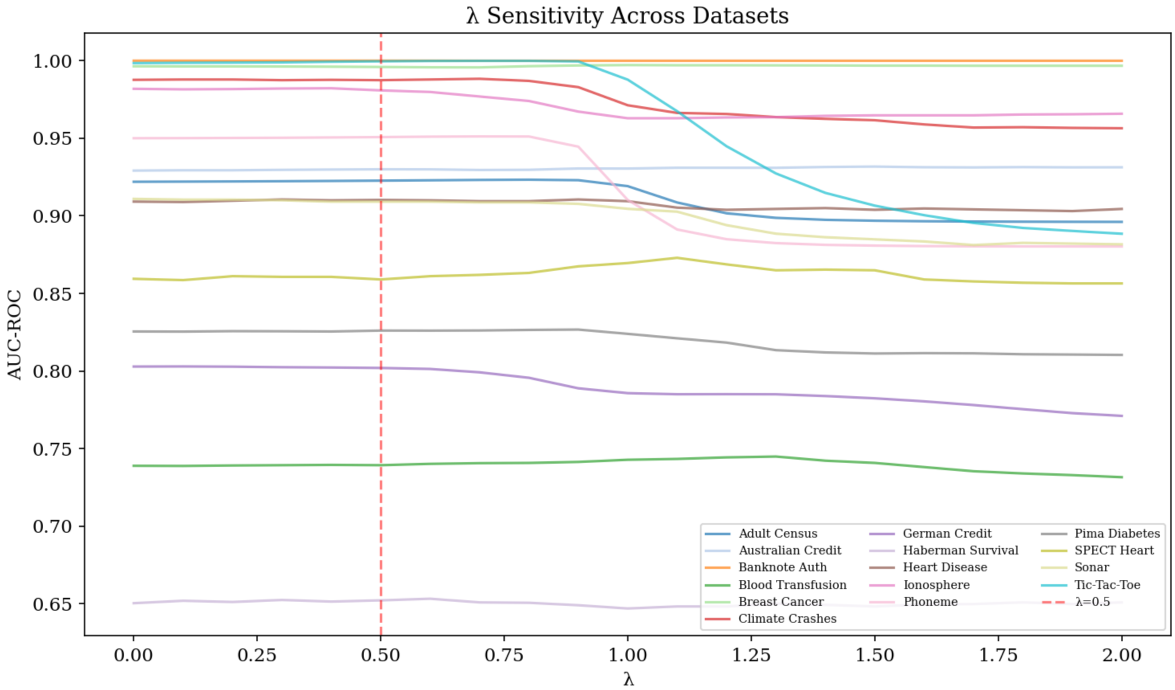

Figure 4.7: AUC-ROC as a function of λ for all 15 datasets. The red dashed line marks the proposed default $\lambda = 0.5$. Interpretation of the three dataset-difficulty patterns visible in the figure is provided in the body text.

Table 4.4: Summary of hypothesis testing results.

Hypothesis	**Result**	**Key Evidence**
H1: Adding interaction information improves performance	Not significant ($p = 0.148$)	Wins on 9/15 datasets, $+0.0018$ mean AUC, best Brier score among ablation variants
H2: Dividends accurately detect model relationships	Partially supported	3/4 controlled tests pass; negative control fails criterion 1 due to finite-sample AUC noise
H3: Framework is competitive with standard methods	Not significant ($p = 0.072$)	Ranks 2nd of 6 methods; best calibration
H4: A fixed $\lambda = 0.5$ works well everywhere	Not supported	Optimal λ ranges from 0.0 to 1.5; cross-validation needed

CHAPTER 5

DISCUSSION

5.1 Interpretable Diagnostics as the Primary Contribution

The pairwise dividend matrix offers something that no other ensemble method provides: a pairwise, interpretable map of which models overlap and which are most distinct. From the matrix, a practitioner can immediately identify the most redundant pair and the least redundant pair in the pool. If budget constraints require dropping a model, the matrix directly identifies candidates whose removal would incur the least loss of complementarity. If a new model is being considered for the pool, its pairwise dividends with existing members immediately reveal whether it adds genuine diversity or merely more redundancy. Stacking, equal weighting, and performance-proportional weighting offer no comparable diagnostic capability.

This interpretability benefit is independent of whether the coopetitive weights themselves improve upon simpler baselines, and it constitutes the framework's most durable practical contribution. The diagnostic value is also directly applicable to automated machine learning pipelines, ensemble pruning tools, and any setting where interpretable model relationship information is valued alongside raw predictive accuracy. The quantitative performance results reported in §§ 4.3–4.4 should be read as secondary evidence: they characterize the magnitude of any performance trade-off relative to simpler baselines, not the primary reason to adopt the framework.

However, the results also reveal an important limitation of the "coopetitive" framing. In theory, the framework encompasses both a cooperative regime (where positive dividends indicate complementarity) and a competitive regime (where negative dividends indicate redundancy). In practice, all pairwise dividends across all fifteen datasets and five random seeds (75 runs in total) were negative, meaning the cooperative half of the coopetitive duality never appeared in this study's setting. Positive dividends would more likely arise from weaker or more diverse base models (e.g., models trained on different feature subsets or data modalities), from small or noisy datasets where different models make genuinely different errors, or from settings where some models are deliberately weak. In the present setting of five well-optimized classifiers on standard tabular binary-classification benchmarks, the framework functioned entirely as a redundancy-penalty mechanism. The term "coopetitive" remains aspirationally apt, describing the potential dual character of the formula, but practitioners should be aware that in well-optimized heterogeneous pools the competitive component will dominate.

5.2 Role of the Interaction Term

The interaction term in the coopetitive formula functions primarily as a redundancy penalty. Because nearly all pairwise dividends are negative across all datasets, increasing λ effectively downweights models that overlap heavily with many others and upweights models that are more distinct from the pool. The performance improvement is modest ($+0.0018$ AUC) because the five heterogeneous models in the pool are already reasonably diverse, providing the interaction term limited room to improve an arrangement that is already decent. In pools with greater redundancy, such as multiple gradient boosting variants with different hyperparameter settings, the penalty mechanism is expected to exert a stronger effect and the gain over simpler baselines would be correspondingly larger, though this expectation is untested here. The modest gain observed here should therefore not be interpreted as evidence that the interaction term is unimportant in general, but rather as a consequence of the specific diversity level of the five-model heterogeneous pool used in this study.

5.3 Evidence Consistent with the Double-Counting Hypothesis

One of the clearest findings is that the Shapley + SII variant performed worse than plain Shapley by 0.009 AUC. This result is consistent with the double-counting hypothesis motivating the entire framework: when interaction effects already embedded within the Shapley value are added again through the Shapley Interaction Index, the resulting weights are distorted and performance drops. The Möbius-based framework avoids this by construction, since singleton and pairwise dividends are mathematically non-overlapping. The magnitude of this degradation (0.009 AUC) is meaningful given that the total AUC spread across the ten ablation variants (Table 4.1) is only 0.027, meaning the double-counting cost represents roughly one-third of that spread (from the theoretically motivated Interaction-Only lower bound of 0.8707 to the Coopetitive-0.5 ceiling of 0.8977).

5.4 Why Stacking Underperformed

Stacking is a widely used and often strong baseline, but it ranked fourth here (rank 3.50, AUC $= 0.8852$). The most plausible explanation is sample size. Eleven of the fifteen benchmark datasets have fewer than 1,000 samples, and the 20% validation split gives the meta-learner as few as 40–60 training instances. With five correlated base model predictions as input features and only a few dozen rows to learn from, the logistic regression meta-learner is likely to overfit, a known risk when the number of training instances is small relative to the number of input features. Game-theoretic methods sidestep this problem entirely because they derive weights from the combinatorial structure of 32 coalition values, which does not require fitting parameters to small samples. In larger-sample settings, we conjecture that stacking would outperform game-theoretic methods given larger training samples, though this conjecture is untested here.

5.5 The Decomposition Coverage Issue

Coverage fell far below the 80% threshold specified in the diagnostic methodology (§ 3.4.2). The universally negative values (-2.6 to -5.0) are a direct consequence of truncating the Möbius decomposition to only two orders. In a pool of five strong, partially overlapping classifiers, the ten pairwise redundancy terms overwhelm the five positive singleton terms. The missing higher-order terms (10 three-way, 5 four-way, and 1 five-way interaction) would restore the balance, but computing and interpreting them is beyond the scope of this study. The practical takeaway is that this metric should not be read as a completeness measure but as a diagnostic flag: strongly negative values indicate a pool dominated by pairwise redundancy, signaling that the pool would benefit from either a more diverse model selection or explicit higher-order interaction terms in the weighting formula.

5.6 Why λ Cannot Be Fixed

The optimal λ varies from 0.0 (for easy datasets where interaction reweighting adds only noise) to 1.5 (for difficult datasets where aggressive redundancy penalization helps). This makes intuitive sense: when every model in the pool already achieves near-perfect accuracy, there is very little redundancy to penalize, and adjusting weights based on interaction structure introduces noise rather than signal. When models struggle and overlap substantially in their errors, a larger λ more forcefully redirects weight toward the most distinct models, yielding a more robust ensemble. The practical consequence is that the framework should always be deployed with cross-validated λ selection. This adds almost no computational cost because the inner cross-validation operates on cached probability predictions and no models need to be retrained.

5.7 Grand Coalition Is Not Always Best

In 92% of experimental runs, some subset of models outperformed the full five-model ensemble. This is not a flaw in the framework but a reflection of the redundancy in the pool: averaging five partially overlapping models can be worse than combining just the three or four most distinct ones. The dividend matrix provides exactly the information needed to identify which models to retain and which to exclude, suggesting that future work should combine the coopetitive weighting mechanism with a dividend-informed pruning step to further improve performance by first excluding the most redundant models before applying ensemble combination.

CHAPTER 6

CONCLUSION

6.1 Summary of Findings

This study set out to address three gaps in ensemble weighting: entanglement of individual and interaction effects in the Shapley value, double-counting when Shapley values are augmented with the SII, and the absence of an interpretable pairwise diagnostic. The Möbius decomposition resolves the interpretability gap most clearly: the pairwise dividend matrix provides a pairwise map of model redundancy that no other ensemble method produces, directly informing decisions about model selection, pool construction, and ensemble pruning. The double-counting gap is resolved empirically as well as theoretically: the Shapley + SII variant degraded performance by 0.009 AUC relative to plain Shapley, roughly one-third of the total AUC spread across all ablation variants, confirming that the non-overlapping property of Harsanyi dividends matters in practice. The entanglement gap was not resolved in a statistically significant way (H1: $p = 0.148$); while the framework ranked competitively in the benchmark, H3 also did not reach conventional significance ($p = 0.072$), though both showed directional consistency with their respective hypotheses.

Construct validity was partially established: three of four controlled tests confirmed that pairwise dividends correctly detect redundancy, agree with established interaction measures ($r = 0.964$ with the SII), and respond smoothly to changing model similarity. Test D failed criterion 1 (maximum absolute pairwise dividend 0.039 exceeded the 0.01 threshold) but satisfied criterion 2; the excess is attributed to finite-sample AUC noise with approximately 160 validation samples. In the benchmark comparison, Coopetitive-CV ranked second among six methods. Separately, Coopetitive-0.5 achieved the best probability calibration of any ablation variant tested (Neg. Brier $= -0.097$).

6.2 Conclusions

The primary contribution of this study is the pairwise dividend matrix as a troubleshooting instrument for AUC – a diagnostic layer between model selection and ensemble performance that no existing weighting method provides. The relationship between the matrix and AUC is direct and actionable. When all pairwise dividends are strongly negative, as observed across all fifteen datasets here, the pool is governed by redundancy: the ensemble cannot leverage complementarity to improve AUC, and no re-weighting of redundant models will recover the lost performance. In this regime, the matrix identifies the highest-leverage intervention – replace or exclude the model pair with the most negative dividend – rather than leaving the practitioner to adjust weights among equally overlapping models by trial and error. Conversely, when a pair carries a less negative or positive dividend, it is providing genuine complementarity; preserving or upweighting that pairing is the action most likely to push AUC upward.

The matrix also provides a principled basis for setting λ, the parameter that mediates between the interaction structure and the final ensemble weights. Because the optimal λ ranged from 0.0 (pools where redundancy is negligible and interaction penalization adds only noise) to 1.5 (pools where aggressive penalization of overlapping models is needed), dividend magnitude functions as a natural prior for this choice: a pool with dividends near zero warrants $\lambda \approx 0$, while one with dividends averaging -0.3 or below warrants higher λ. This connection closes the loop from diagnosis to action – inspect the pairwise matrix, read the redundancy profile, set λ accordingly, and observe the downstream AUC effect. For the field of ensemble learning, this constitutes a formal, interpretable bridge between pairwise model interaction structure and AUC optimization that performance-focused methods like stacking, regardless of their accuracy, do not share.

6.3 Limitations

Three limitations proved most consequential in practice. First, the decomposition coverage was universally and strongly negative (ranging from -2.58 to -4.98), indicating that truncating at pairwise terms misses substantial higher-order interaction structure; including three-way and above dividends is the most direct extension. Second, the optimal λ varied widely across datasets (0.0 to 1.5), ruling out a universal fixed default and requiring cross-validated selection for deployment. Third, the rank-stability diagnostic showed perfect Spearman correlation across all 75 runs, confirming that model rankings are preserved under the framework's weighting scheme; however, with only $n = 5$ models this diagnostic has limited power, and a larger pool is needed to make this check meaningful. Additional boundaries include the restriction to binary classification and the exponential scaling of exact coalition evaluation ($n \leq 12$).

6.4 Recommendations and Future Work

(a) Extensions. The most direct extension is to include three-way and higher-order Harsanyi dividends in the weighting formula, which would restore full decomposition coverage and address the framework's largest technical limitation. Generalizing to multi-class classification and regression settings would broaden applicability substantially, as would scaling to larger model pools via approximation schemes (e.g., sampling-based estimation of coalition values) that would remove the $n \leq 12$ constraint.

(b) Refinements. The pairwise dividend matrix could be integrated with ensemble pruning by using it to identify and exclude the most redundant models before applying ensemble combination, potentially recovering performance lost to the grand coalition suboptimality observed in 92% of experimental runs. The softmax normalization scheme used to derive weights should be compared against simpler clamping-and-renormalization approaches, and alternative sharing factors beyond the $\frac{1}{2}$ Shapley sharing rule deserve exploration. Larger model pools ($n > 5$) are needed to give the rank-stability diagnostic sufficient discriminating power.

(c) Applications. Online and streaming scenarios are a longer-term direction. The primary bottleneck is recomputing 2^n coalition values incrementally as data arrives, which requires approximation methods beyond the scope of this study. The framework is also a natural fit for automated machine learning pipelines, where the dividend matrix could inform pool construction and model selection in a principled, interpretable way without requiring manual inspection. Rather than claiming that the coopetitive framework will necessarily outperform stacking in raw predictive accuracy, the contribution lies in offering interpretable diagnostics

that more powerful but opaque methods do not provide.

Data and Code Availability

The code and data used in this study are publicly available at `https://github.com/maxellmilay/mobius-decomposition-ensemble-weighting`.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude and Claude Code (Anthropic) in order to assist with code development, review sections of the paper for clarity and structure, and paraphrase selected passages. The author reviewed and edited all AI-assisted content as needed and takes full responsibility for the content of the submitted manuscript.

BIBLIOGRAPHY

Bimonte, G., Russolillo, M., Shang, H. L., & Yang, Y. (2024). Mortality models ensemble via Shapley value. *Decisions in Economics and Finance*. `https://doi.org/10.1007/s10203-024-00455-z`

Brandenburger, A. M., & Nalebuff, B. J. (1996). *Co-opetition*. Doubleday.

Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24(2), 123–140.

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.

Dehez, P. (2017). On Harsanyi dividends and asymmetric values. *International Game Theory Review*, 19(3), 1–36. `https://doi.org/10.1142/S0219198917500128`

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7, 1–30.

Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874.

Freund, Y., & Schapire, R. E. (1997). A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1), 119–139.

Grabisch, M., & Roubens, M. (1999). An axiomatic approach to the concept of interaction among players in cooperative games. *International Journal of Game Theory*, 28(4), 547–565.

Harsanyi, J. C. (1959). A bargaining model for cooperative n-person games. In A. W. Tucker & R. D. Luce (Eds.), *Contributions to the Theory of Games IV* (pp. 325–355). Princeton University Press.

Harsanyi, J. C. (1963). A simplified bargaining model for the n-person cooperative game. *International Economic Review*, 4(2), 194–220.

Kuncheva, L. I., & Whitaker, C. J. (2003). Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2), 181–207.

Margineantu, D. D., & Dietterich, T. G. (1997). Pruning adaptive boosting. In *Proceedings of the 14th International Conference on Machine Learning* (pp. 211–218).

Muschalik, M., Baniecki, H., Fumagalli, F., Kolpaczki, P., Hammer, B., & Hüllermeier, E. (2024). shapiq: Shapley interactions for machine learning. *NeurIPS 2024, Datasets and Benchmarks Track*. `https://arxiv.org/abs/2410.01649`

Rozemberczki, B., & Sarkar, R. (2021). The Shapley value of classifiers in ensemble games. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management* (pp. 1558–1567).

Rozemberczki, B., Watson, L., Bayer, P., Yang, H.-T., Kiss, O., Nilsson, S., & Sarkar, R. (2022). The Shapley value in machine learning. In *Proceedings of IJCAI-22* (pp. 5572–5579).

Shapley, L. S. (1953). A value for n-person games. In *Contributions to the Theory of Games II* (pp. 307–317). Princeton University Press.

Terpilowski, M. (2019). scikit-posthocs: Pairwise multiple comparison tests in Python. *Journal of Open Source Software*, 4(36), 1169.

Tsai, C.-P., Yeh, C.-K., & Ravikumar, P. (2023). Faith-Shap: The faithful Shapley interaction index. *Journal of Machine Learning Research*, 24(94), 1–42.

Zhou, Z.-H. (2012). *Ensemble Methods: Foundations and Algorithms*. CRC Press.

Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.